

Electrical Circuits I

ENGR 2405

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Chapter 09 – Sinusoids and Phasors

Chapter 10 – Sinusoidal Steady-State Analysis

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Overview

- This lecture will cover alternating current
- Complex numbers will be introduced
- Applications of phasors and frequency domain analysis for circuits including resistors, capacitors, and inductors will be covered
- Circuit analysis will be (re-)introduced
- NA, MA, Superposition, & Source Transformation will be covered
- Applications in op-amps and oscillators are reviewed

Alternating Current

- Alternating Current, or AC, is the dominant form of electrical power that is delivered to homes and industry
- In the late 1800s there was a battle between proponents of DC and AC
- AC won out due to its efficiency for long distance transmission
- AC is a sinusoidal current, meaning the current reverses at regular times and has alternating positive and negative values

Sinusoids

- Sinusoids are interesting to us because there are a number of natural phenomenon that are sinusoidal in nature
- It is also a very easy signal to generate and transmit
- Also, through Fourier analysis, any practical periodic function can be made by adding sinusoids
- Lastly, they are very easy to handle mathematically

Sinusoids

- A sinusoidal forcing function produces both a transient and a steady state response
- When the transient has died out, we say the circuit is in sinusoidal steady state
- A sinusoidal voltage may be represented as:

$$v(t) = V_m \sin \omega t$$

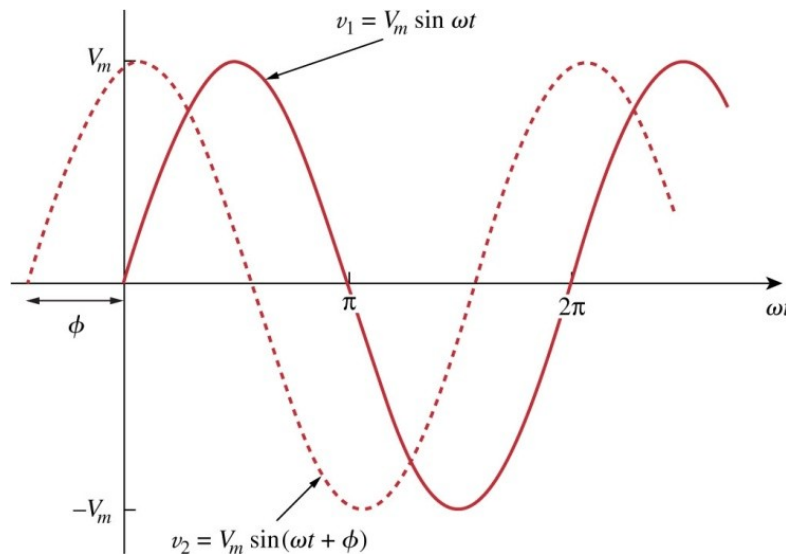
- One characteristic of a sinusoidal function is that it repeats itself every T seconds (period) or has a frequency f (Hz):

$$T = \frac{2\pi}{\omega} \quad f = \frac{1}{T}$$

Sinusoids

- More generally, we need to account for relative timing of one wave versus another
- This can be done by including a phase shift, ϕ :
- Consider the two sinusoids:

$$v_1(t) = V_m \sin(\omega t) \text{ and } v_2(t) = V_m \sin(\omega t + \phi)$$



Complex Numbers

- A powerful method for representing sinusoids is the phasor
- But in order to understand how they work, we need to cover some complex numbers first
- A complex number z can be represented in rectangular form as:

$$z = x + jy$$

- It can also be written in polar or exponential form as:

$$z = r \angle \phi = r e^{j\phi}$$

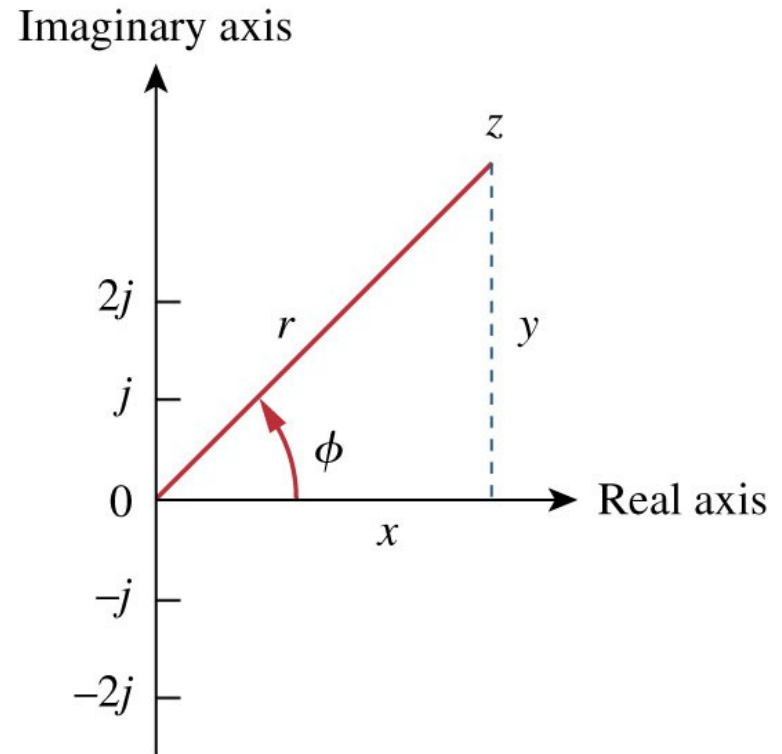
Complex Numbers

- The different forms can be inter-converted
- Starting with rectangular form, one can go to polar:

$$r = \sqrt{x^2 + y^2} \quad \phi = \tan^{-1} \frac{y}{x}$$

- Likewise, from polar to rectangular form goes as follows:

$$x = r \cos \phi \quad y = r \sin \phi$$



Complex Numbers

- The following mathematical operations are important.

Addition

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication

$$z_1 z_2 = r_1 r_2 \angle (\phi_1 + \phi_2)$$

Division

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle (\phi_1 - \phi_2)$$

Reciprocal

$$\frac{1}{z} = \frac{1}{r} \angle (-\phi)$$

Square Root

$$\sqrt{z} = \sqrt{r} \angle (\phi / 2)$$

Complex Conjugate

$$z^* = x - jy = r \angle -\phi = r e^{-j\phi}$$

Practice Problems

9.3 Evaluate the following complex numbers:

a. $[(5 + j2)(-1 + j4) - 5 \angle 60^\circ]^*$

b. $\frac{10 + j5 + 3 \angle 60^\circ}{-3 + j4} + 10 \angle 30^\circ + j5$

Phasors

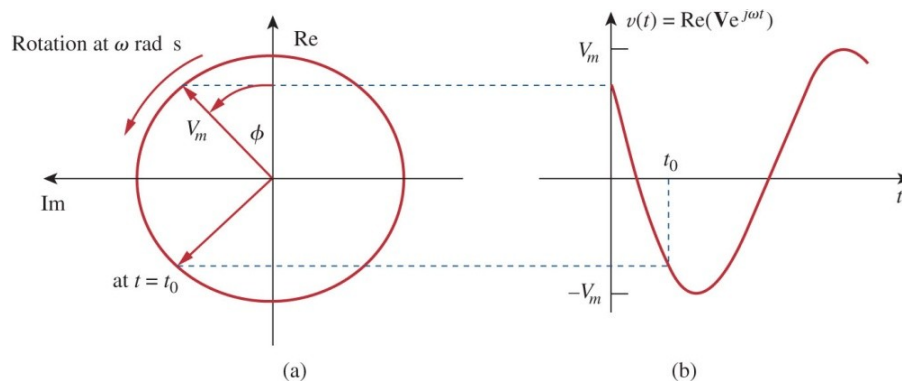
- The idea of a phasor representation is based on Euler's identity:
$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$
- From this we can represent a sinusoid as the real component of a vector in the complex plane
- The length of the vector is the amplitude of the sinusoid
- The vector, V , in polar form, is at an angle ϕ with respect to the positive real axis

Phasors

- Phasors are typically represented at $t = 0$
- As such, the transformation between time domain to phasor domain is:

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & V = V_m \angle \phi \\ \text{(time-domain)} & & \text{(phasor-domain)} \end{array}$$

- They can be graphically represented as shown here



Sinusoid-Phasor Transformation

- Note that the frequency of the phasor is not explicitly shown in the phasor diagram
- For this reason phasor domain is also known as frequency domain
- Applying a derivative to a phasor yields:

$$\frac{dv}{dt} \Leftrightarrow j\omega V$$

- Applying an integral to a phasor yields:

$$\int v dt \Leftrightarrow \frac{V}{j\omega}$$

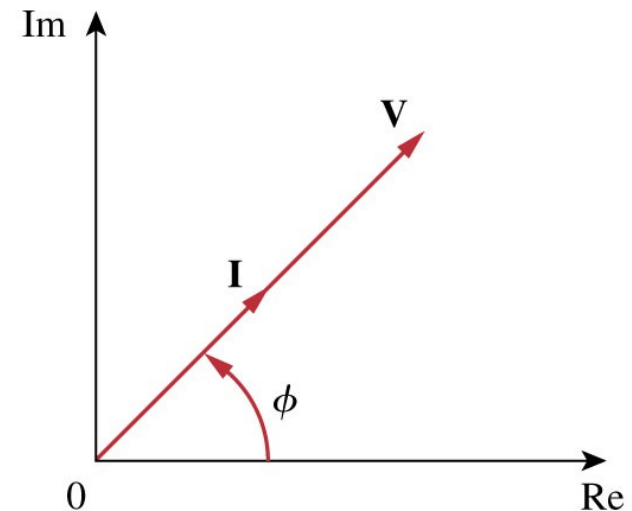
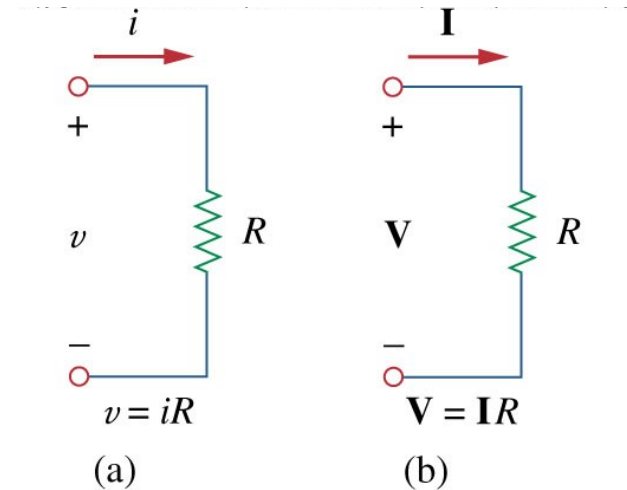
Practice Problems

9.7 Find the voltage $v(t)$ in:

$$2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

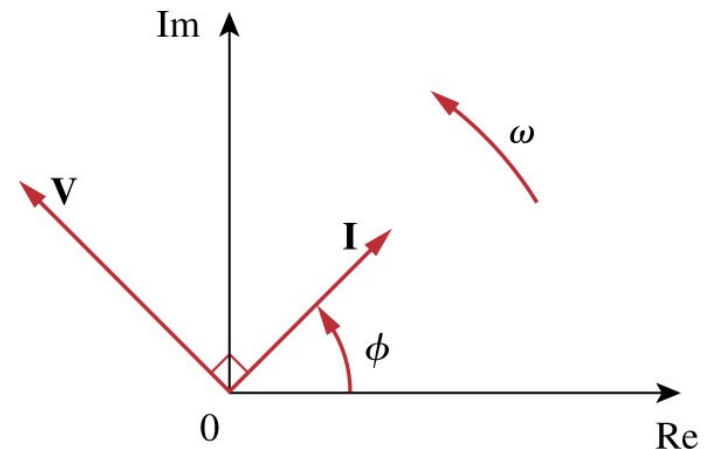
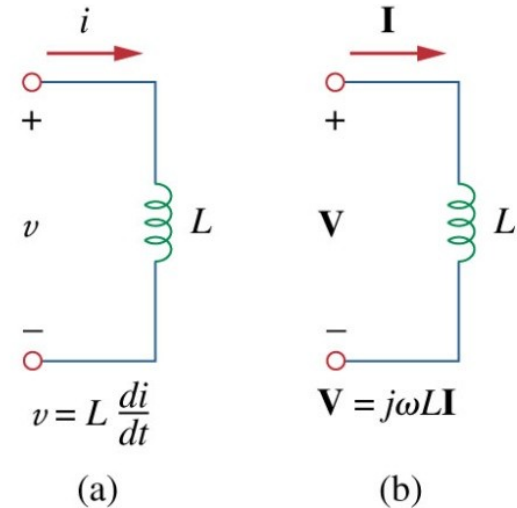
Phasor Relationships for Resistors

- Each circuit element has a relationship between its current and voltage
- These can be mapped into phasor relationships very simply for resistors capacitors and inductor
- For the resistor, the voltage and current are related via Ohm's law
- As such, the voltage and current are in phase with each other



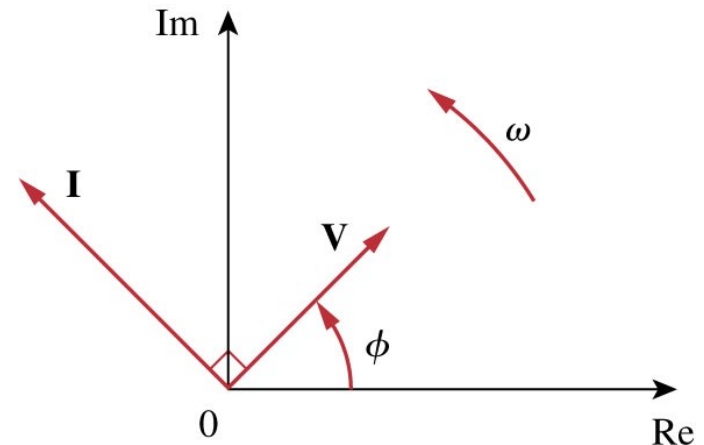
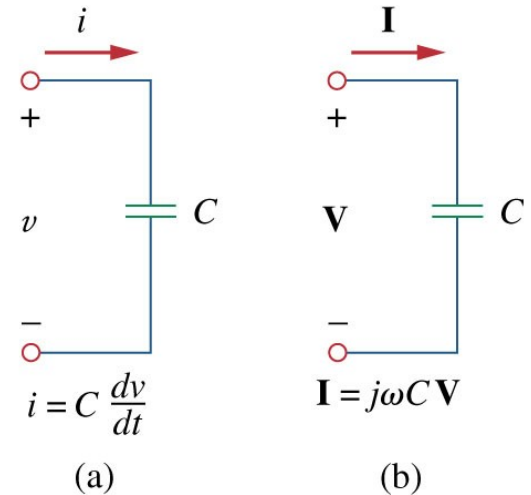
Phasor Relationships for Inductors

- Inductors on the other hand have a phase shift between the voltage and current
- In this case, the voltage leads the current by 90°
- Or one says the current lags the voltage, which is the standard convention
- This is represented on the phasor diagram by a positive phase angle between the voltage and current



Phasor Relationships for Capacitors

- Capacitors have the opposite phase relationship as compared to inductors
- In their case, the current leads the voltage
- In a phasor diagram, this corresponds to a negative phase angle between the voltage and current



Voltage current relationships

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C \frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

Practice Problems

9.8 If:

$$v = 10 \cos(100t + 30^\circ)$$

is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the cap

Impedance and Admittance

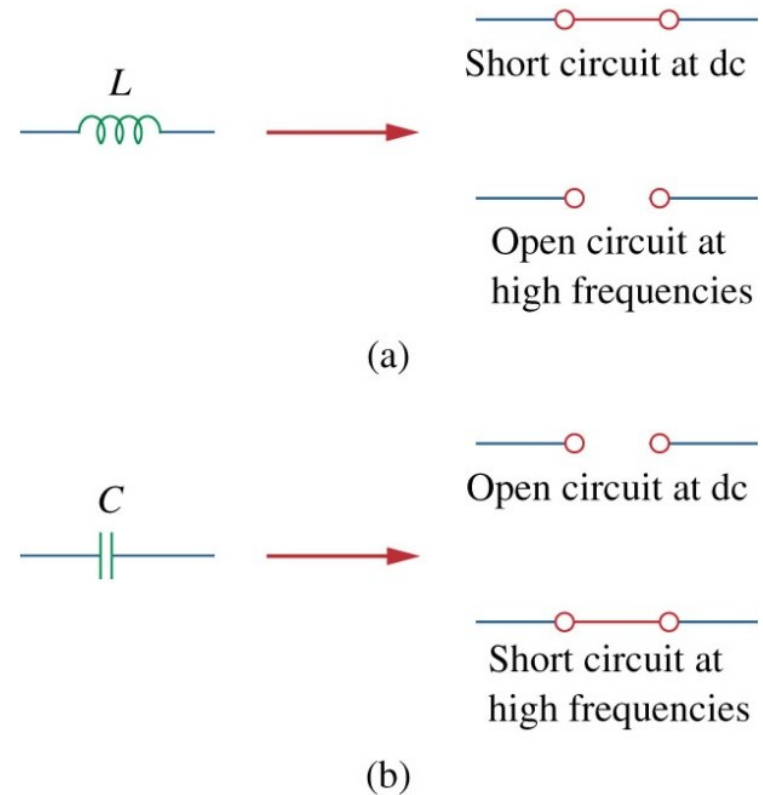
- It is possible to expand Ohm's law to capacitors and inductors
- In time domain, this would be tricky as the ratios of voltage and current are always changing
- But in frequency domain it is straightforward
- The impedance of a circuit element is the ratio of the phasor voltage to the phasor current

$$Z = \frac{V}{I} \quad \text{or} \quad V = IZ$$

- Admittance is simply the inverse of impedance

Impedance and Admittance

- It is important to realize that in frequency domain, the values obtained for impedance are only valid at that frequency
- Changing to a new frequency will require recalculating the values
- The impedance of capacitors and inductors are shown here



Impedance and Admittance

- As a complex quantity, the impedance may be expressed in rectangular form
- The separation of the real and imaginary components is useful
- The real part is the resistance
- The imaginary component is called the reactance, X
- When it is positive, we say the impedance is inductive, and capacitive when it is negative

Impedance and Admittance

- Admittance, being the reciprocal of the impedance, is also a complex number
- The real part of the admittance is called the conductance, G
- The imaginary part is called the susceptance, B
- These are all expressed in Siemens or (mhos)
- The impedance and admittance components can be related to each other:

$$G = \frac{R}{R^2 + X^2} \quad B = -\frac{X}{R^2 + X^2}$$

Impedance and Admittance

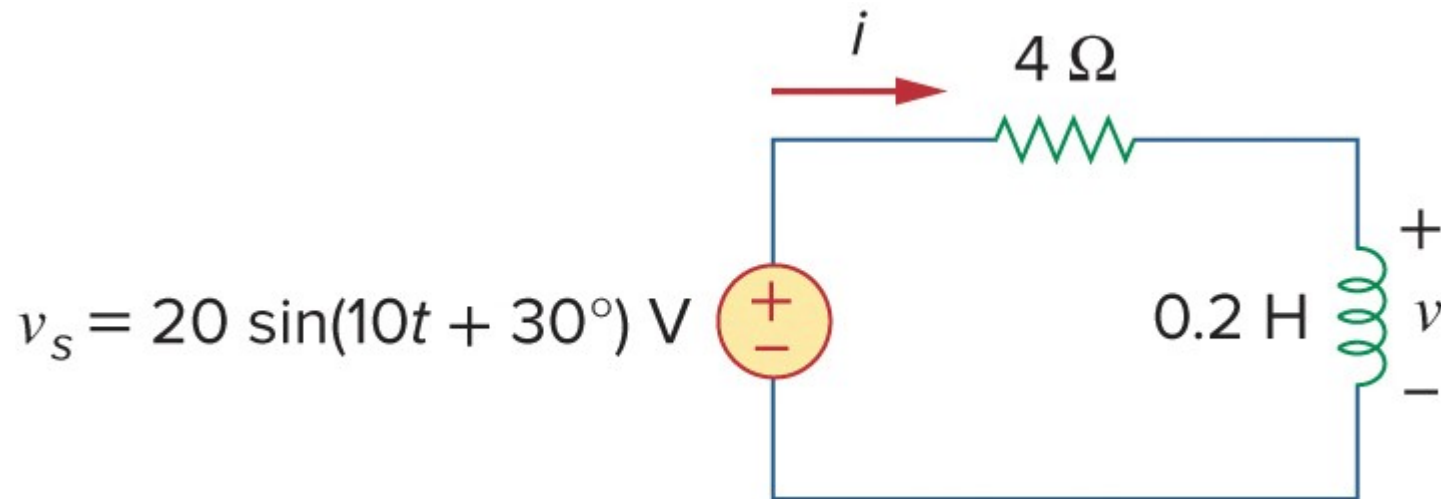
TABLE 9.3

Impedances and admittances of passive elements.

Element	Impedance	Admittance
R	$Z = R$	$Y = \frac{1}{R}$
L	$Z = j\omega L$	$Y = \frac{1}{j\omega L}$
C	$Z = \frac{1}{j\omega C}$	$Y = j\omega C$

Practice Problems

9.9 Determine $v(t)$ and $i(t)$

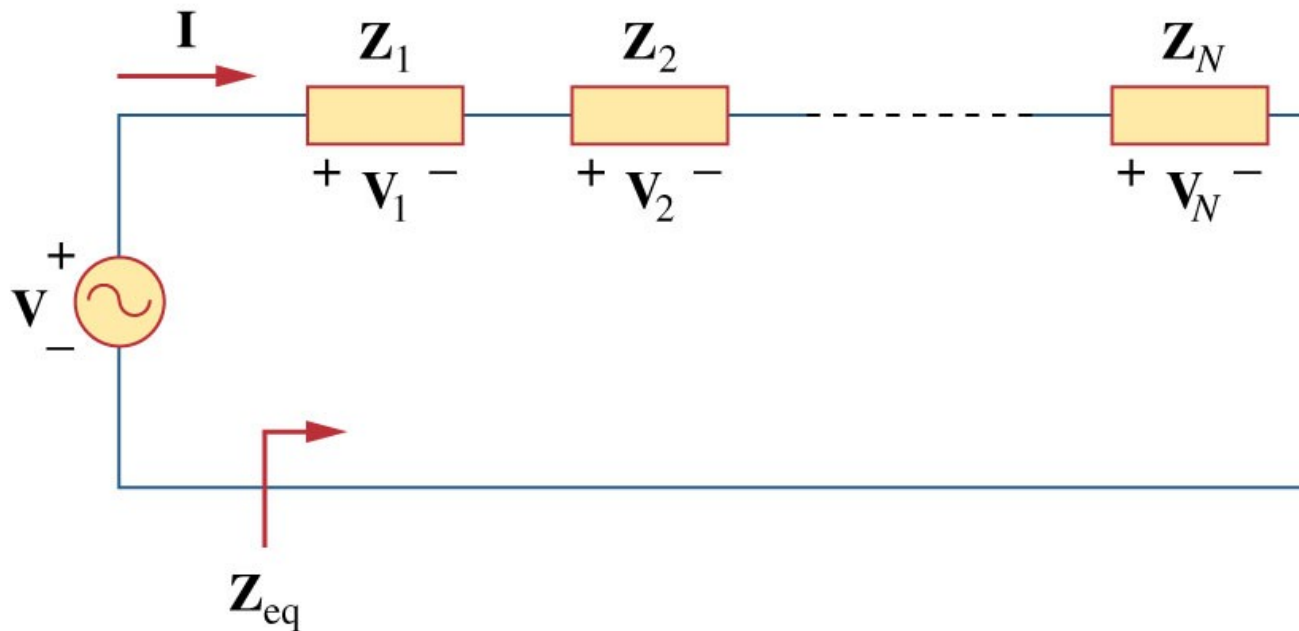


Kirchoff's Laws in Frequency Domain

- A powerful aspect of phasors is that Kirchoff's laws apply to them as well!
- This means that a circuit transformed to frequency domain can be evaluated by the same methodology developed for KVL and KCL
- One consequence is that there will likely be complex values

Impedance Combinations

- Once in frequency domain, the impedance elements are generalized
- Combinations will follow the rules for resistors:



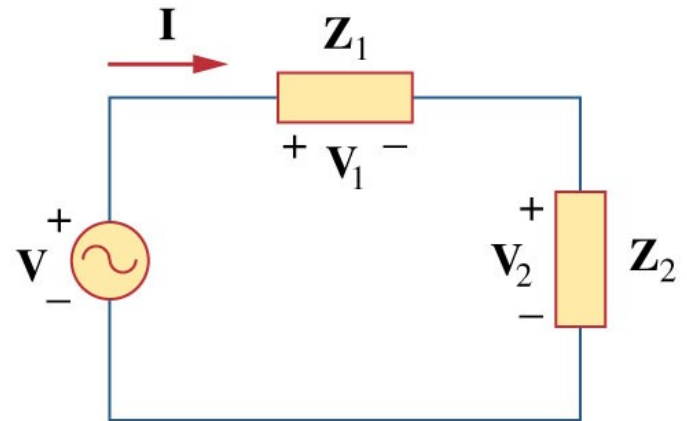
Impedance Combinations

- Series combinations will result in a sum of the impedance elements:

$$Z_{eq} = Z_1 + Z_2 + \cdots + Z_N$$

- Here then two elements in series can act like a voltage divider.

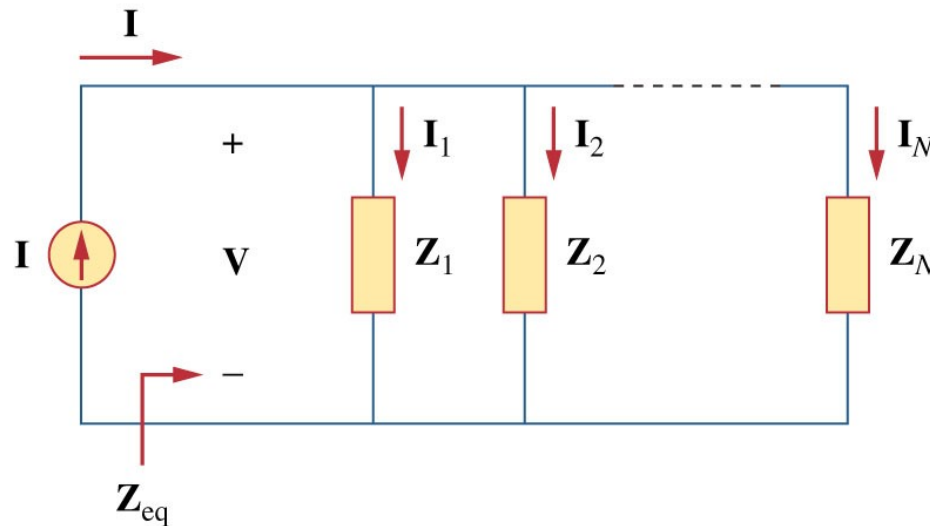
$$V_1 = \frac{Z_1}{Z_1 + Z_2} V \quad V_2 = \frac{Z_2}{Z_1 + Z_2} V$$



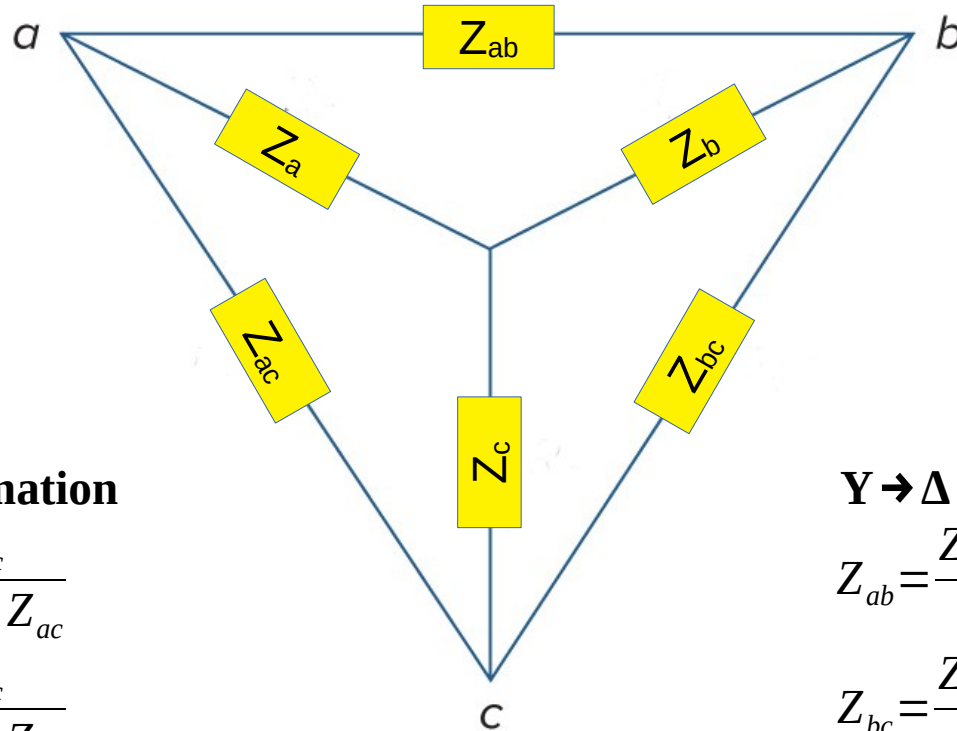
Parallel Combination

- Likewise, elements combined in parallel will combine in the same fashion as resistors in parallel:

$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \dots + \frac{1}{Z_N}$$



Delta-Wye Transformation



$\Delta \rightarrow Y$ Transformation

$$Z_a = \frac{Z_{ab} Z_{ac}}{Z_{ab} + Z_{bc} + Z_{ac}}$$

$$Z_b = \frac{Z_{ab} Z_{bc}}{Z_{ab} + Z_{bc} + Z_{ac}}$$

$$Z_c = \frac{Z_{ac} Z_{bc}}{Z_{ab} + Z_{bc} + Z_{ac}}$$

$Y \rightarrow \Delta$ Transformation

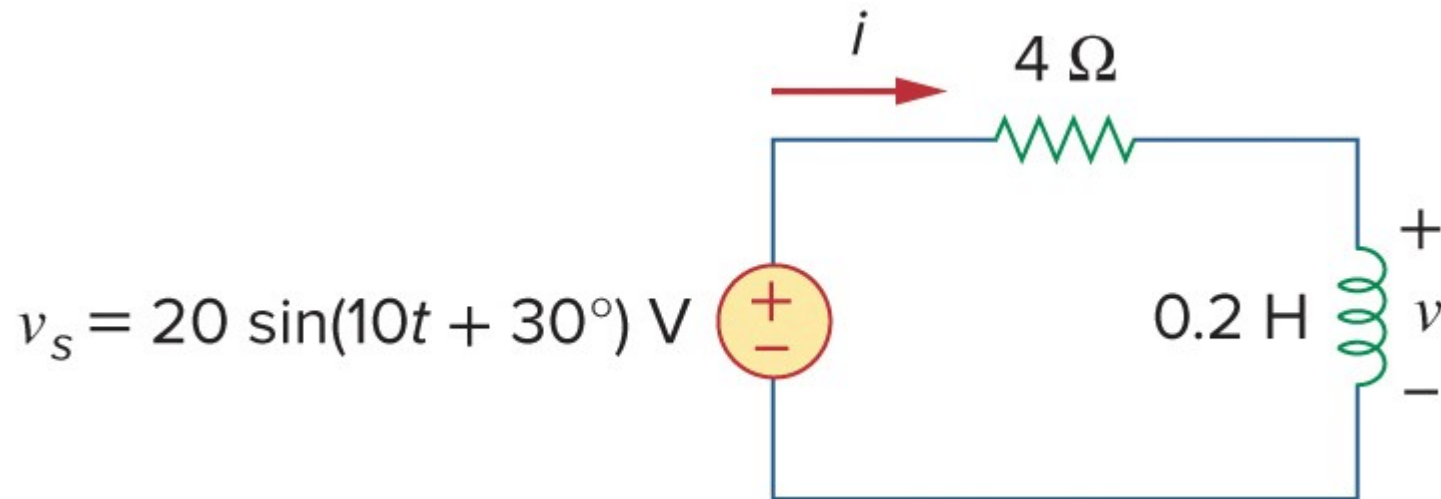
$$Z_{ab} = \frac{Z_a Z_b + Z_b Z_c + Z_a Z_c}{Z_c}$$

$$Z_{bc} = \frac{Z_a Z_b + Z_b Z_c + Z_a Z_c}{Z_a}$$

$$Z_{ac} = \frac{Z_a Z_b + Z_b Z_c + Z_a Z_c}{Z_b}$$

Practice Problems

9.10 Determine $v(t)$ and $i(t)$

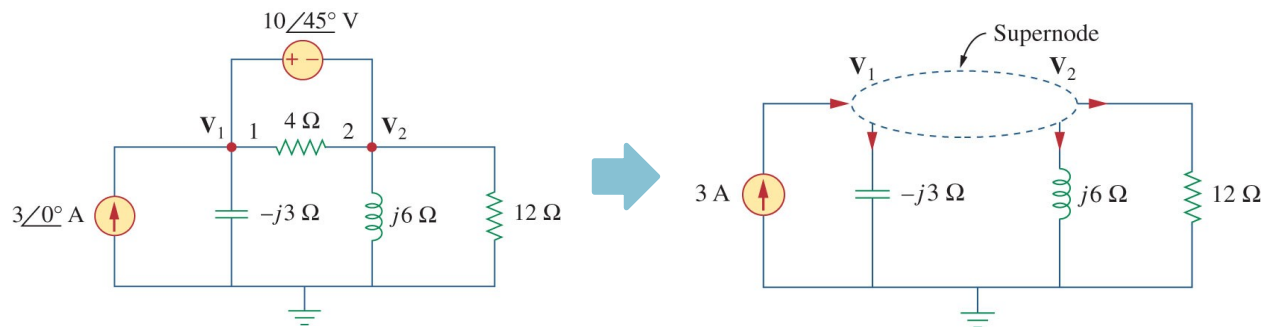


Steps to Analyze an AC Circuit

- There are three steps to analyzing an AC circuit
 - They make use of the fact that frequency domain analysis is simpler because it can make use of the nodal and mesh techniques developed for DC
1. Transform the circuit to the phasor or frequency domain
 2. Solve the problem using circuit techniques
 3. Transform back to time domain

Nodal Analysis

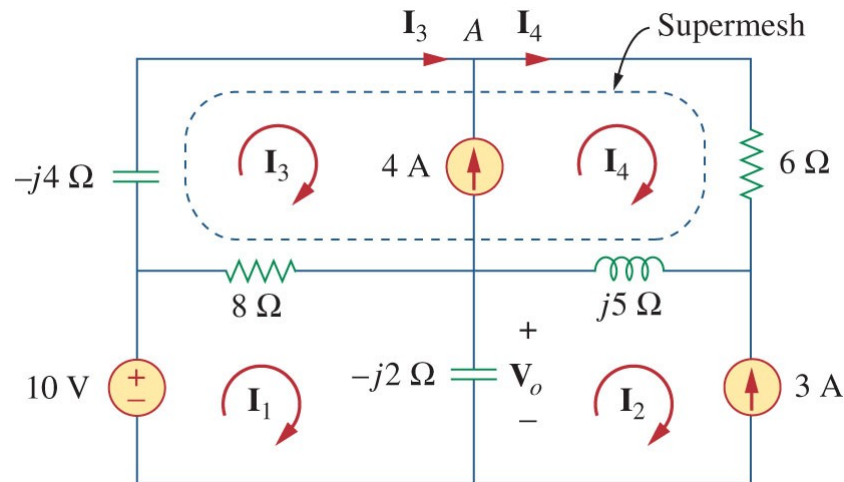
- It is possible to use KCL to analyze a circuit in frequency domain
- Impedances will be expressed as complex numbers
- AC sources appear as DC sources with their values expressed as their amplitude
- The equivalency of the frequency domain treatment compared to the DC circuit analysis includes the use of supernodes



- Note: while complex values will be calculated, all other treatments are the same

Mesh Analysis

- Just as in KCL, the KVL analysis also applies to phasor and frequency domain circuits
- The same rules apply
- In KVL, supermesh analysis is also valid



Superposition

- Since AC circuits are linear, it is also possible to apply the principle of superposition
- This becomes particularly important is the circuit has sources operating at different frequencies
- The complication is that each source must have its own frequency domain equivalent circuit
- Also, the resulting voltages and current must be converted back to time domain before being added

Transformation, Thevenin, & Norton

- Source transformation in frequency domain involves transforming a voltage source in series with an impedance to a current source in parallel with an impedance
- Both Thevenin and Norton's theorems are applied to AC circuits the same way as DC
- The only difference is the fact that the calculated values will be complex

Op Amp AC Circuits

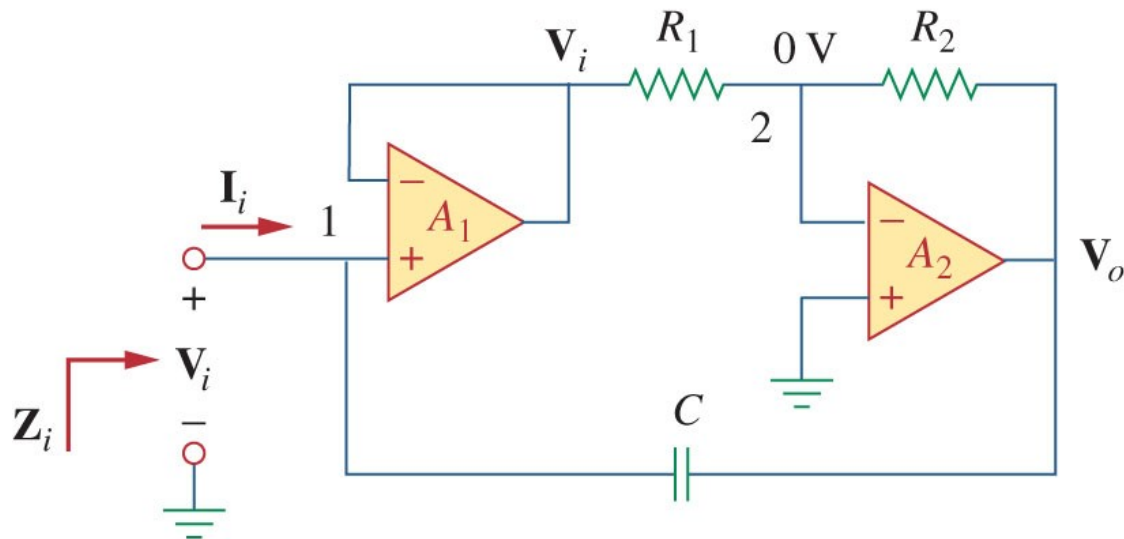
As long as the op amp is working in the linear range, frequency domain analysis can proceed just as it does for other circuits

It is important to keep in mind the two qualities of an ideal op amp:

- No current enters either input terminals
- The voltage across its input terminals is zero with negative feedback

Application

- The op amp circuit shown here is known as a capacitance multiplier
- It is used in integrated circuit technology to create large capacitances out of smaller ones



Capacitor Multiplier

- The first op amp is acting as a voltage follower, while the second one is an inverting amplifier.

- At node 1:

$$I_i = \frac{V_i - V_o}{1/j\omega C} = j\omega C(V_i - V_o)$$

- At node 2:

$$V_o = -\frac{R_2}{R_1} V_i$$

- Combining gives:

$$\frac{I_i}{V_i} = j\omega \left(1 + \frac{R_2}{R_1}\right) C = \frac{1}{Z_i} = j\omega C_{eq} \Rightarrow C_{eq} = \left(1 + \frac{R_2}{R_1}\right) C$$

- Thus, by selecting the resistors, the circuit can produce an effective capacitance between its terminal and ground
- A similar op amp circuit can simulate inductance, eliminating the need to have physical inductors in an IC

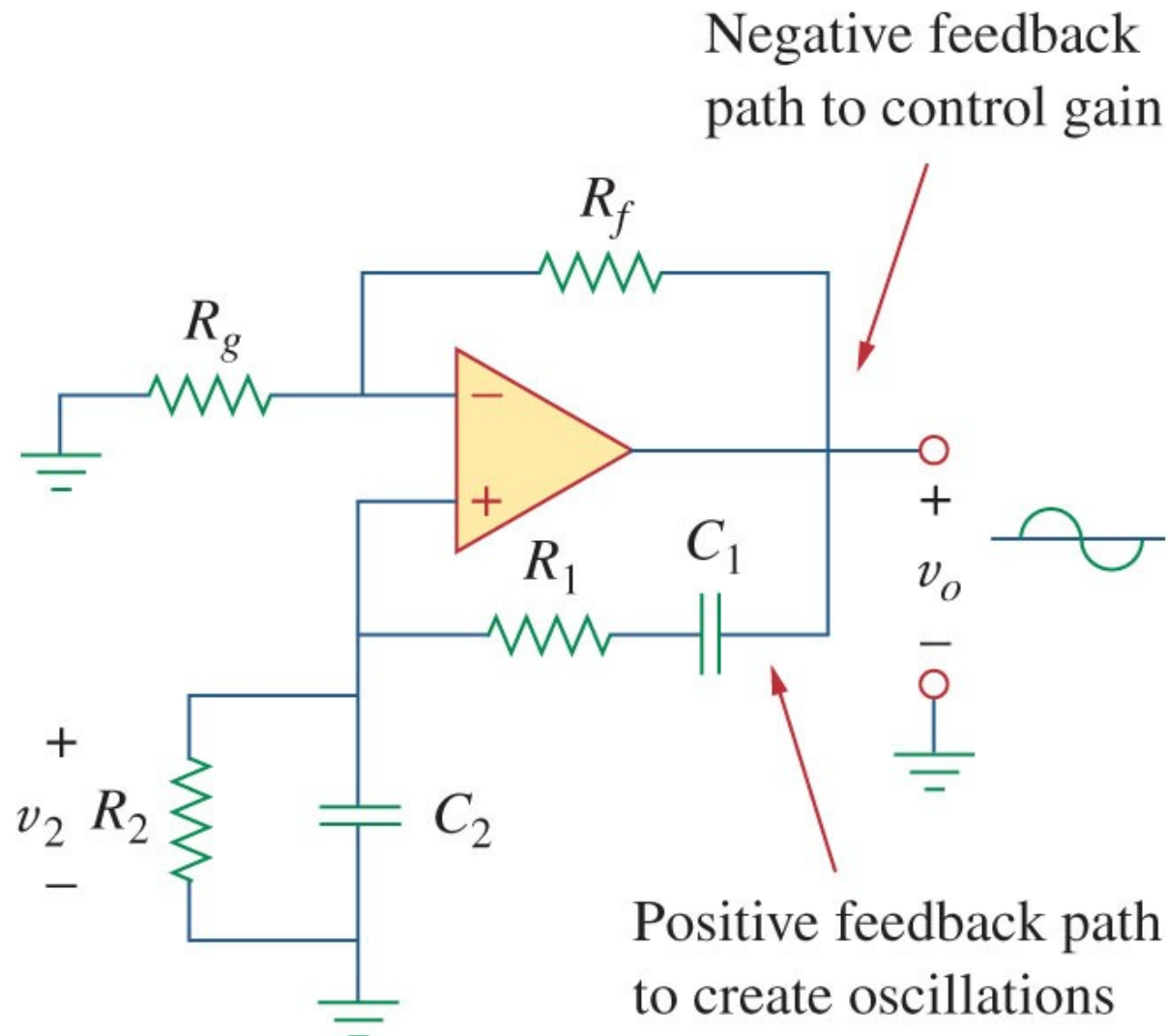
Oscillators

- It is straightforward to imagine a DC voltage source
 - One typically thinks of a battery
- But how to make an AC source?
 - Mains power comes to mind, but that is a single frequency
 - This is where an oscillator comes into play
- They are designed to generate an oscillating voltage at a frequency that is often easily changed

Barkhausen Criteria

- In order for a sine wave oscillator to sustain oscillations, it must meet the Barkhausen criteria:
- Overall gain must be unity or greater (losses must be compensated for by an amplifying device)
- Overall phase shift must be zero
- Three common types of sine wave oscillators are phase-shift, twin T, and Wein-bridge oscillator

Wein-bridge



Wein-bridge

- The circuit is an amplifier in a non-inverting configuration
- There are two feedback paths:
 - The positive feedback path to the non-inverting input creates oscillations
 - The negative feedback path to the inverting input controls the gain
- We can define the RC series and parallel combinations as Z_s and Z_p :

$$Z_s = R_1 - \frac{j}{\omega C_1} \quad Z_p = \frac{R_2}{1 + j\omega R_2 C_2}$$

Wein-bridge

- The resulting feedback ratio is:

$$\frac{V_2}{V_o} = \frac{Z_p}{Z_s + Z_p} = \frac{\omega R_2 C_1}{\omega(R_2 C_1 + R_1 C_1 + R_2 C_2) + j(\omega^2 R_1 C_1 R_2 C_2 - 1)}$$

- To satisfy the 2nd Barkhausen criterion, V_2 must be in phase with V_o , which means this ratio must be real, or:

$$\omega^2 R_1 C_1 R_2 C_2 - 1 = 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$$

- In most cases the capacitors and resistors are made equal, thus the resulting frequency is thus:

$$\omega_0 = \frac{1}{RC} \Rightarrow f_0 = \frac{1}{2\pi RC}$$

Wein-bridge

- Plugging this into the feedback ratio yields:

$$\frac{V_2}{V_o} = \frac{\omega RC}{\omega(RC + RC + RC)} = \frac{1}{3}$$

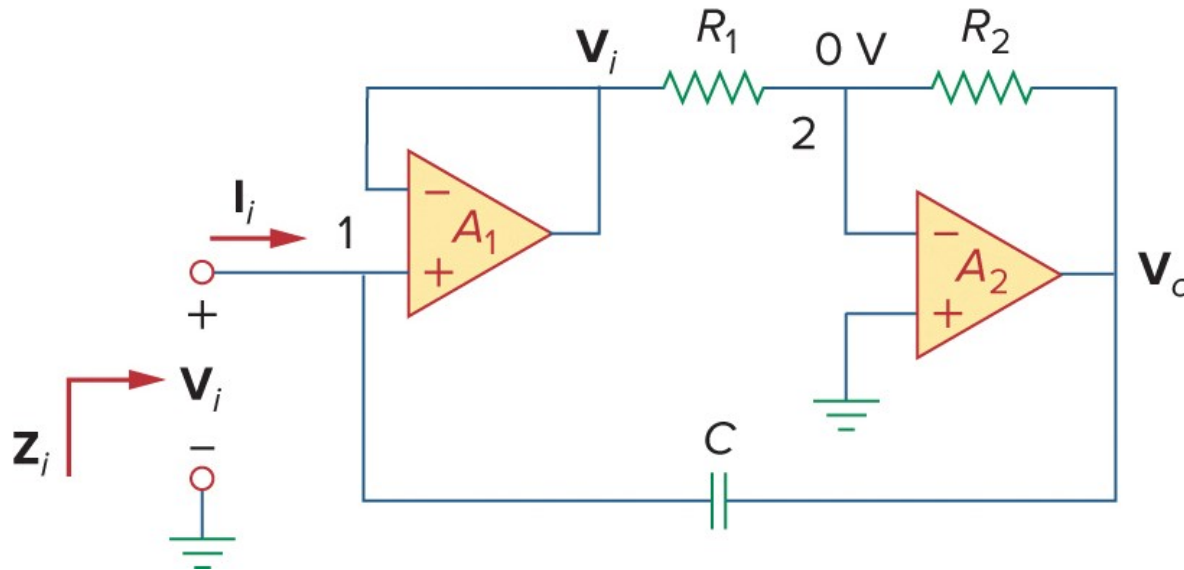
- Thus in order to satisfy the first Barkhausen criteria, the amplifier must provide a gain of 3 or greater.
- Using the feedback equation for a non-inverting amplifier:

$$V_o = \left(1 + \frac{R_f}{R_g}\right) V_2 \Rightarrow \frac{V_o}{V_2} = 1 + \frac{R_f}{R_g} = 3 \Rightarrow$$
$$R_f = 2 R_g$$

Practice Problems

10.15 If: $R_1 = 10\text{ k}\Omega$, $R_2 = 10\text{ M}\Omega$, $C = 10\text{ nF}$

what is the equivalent capacitance in the following circuit?



Practice Problems

10.16 Let: $R_1 = R_2 = 2.5\text{ k}\Omega$, $C_1 = C_2 = 1\text{ nF}$

Determine frequency f_0 of the following oscillator

